

Columbia University
Department of Electrical Engineering
EE E4321. Problem Set #7. Shifter design.
Due: November 18, 2025, 5 PM EST by electronic submission

For Design Project Part II, please submit your writeup as a single PDF file attachment submitted by only one person of your two-person team. Please note clearly in the document the name of the two team members in your group. To submit the layout, please stream out your design according to the instructions on the class website and attach to your submission as well.

Design Project Part II. This problem is part of your final design project; you should work on this with your design-project partner.

You need to design the shifter for your final design project. Your shifter needs to support left logical shifts of 0 to 7 bits using an encoded control line $shift_amount < 0 : 2 >$. I would recommend a logarithmic shifter implementation using three-levels of 2-to-1 multiplexers. You should only have to layout the multiplexer once and reuse this cell 24 times. Watch your bit pitch. Remember that this should match your adder design; ultimately everything in the datapath needs a matched bit pitch.

To complete your shifter design, I expect:

- Sized schematics (make reasonable assumptions about output loading and input capacitance limits)
- Bit-stack layout of your shifter design (remember to begin by making a tiling diagram and then stick layouts of your cells)
- Spectre results for the delay
- Spectre FX runs to verify functionality

Please submit your layout electronically.